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An advanced deep learning method for pepper diseases and pests detection

Xuewei Wang¹, Jun Liu^{1*} and Qian Chen^{2*}

Abstract

Despite the significant progress in deep learning-based object detection, existing models struggle to perform optimally in complex agricultural environments. To address these challenges, this study introduces YOLO-Pepper, an enhanced model designed specifically for greenhouse pepper disease and pest detection, overcoming three key obstacles: small target recognition, multi-scale feature extraction under occlusion, and real-time processing demands. Built upon YOLOv10n, YOLO-Pepper incorporates four major innovations: (1) an Adaptive Multi-Scale Feature Extraction (AMSFE) module that improves feature capture through multi-branch convolutions; (2) a Dynamic Feature Pyramid Network (DFPN) enabling context-aware feature fusion; (3) a specialized Small Detection Head (SDH) tailored for minute targets; and (4) an Inner-CIoU loss function that enhances localization accuracy by 18% compared to standard Ciou. Evaluated on a diverse dataset of 8046 annotated images, YOLO-Pepper achieves state-of-the-art performance, with 94.26% mAP@0.5 at 115.26 FPS, marking an 11.88 percentage point improvement over YOLOv10n (82.38% mAP@0.5) while maintaining a lightweight structure (2.51 M parameters, 5.15 MB model size) optimized for edge deployment. Comparative experiments highlight YOLO-Pepper's superiority over nine benchmark models, particularly in detecting small and occluded targets. By addressing computational inefficiencies and refining small object detection capabilities, YOLO-Pepper provides robust technical support for intelligent agricultural monitoring systems, making it a highly effective tool for early disease detection and integrated pest management in commercial greenhouse operations.

Keywords Pepper diseases and pests detection, Deep learning, YOLOv10, Small object detection, Agricultural scenarios

Introduction

Recent advances in object detection techniques have led to significant improvements across multiple domains, enhancing capabilities in complex target recognition [1], smart IoT-based surveillance [2], and pedestrian group detection in dynamic environments [3]. Moreover,

innovations in optimizing IoT application execution and reinforcing security mechanisms have further boosted real-time computational efficiency and enabled robust detection of adversary behaviors [4, 5]. Although these breakthroughs lay a strong foundation for intelligent systems, their direct application in agricultural disease and pest detection remains challenging, primarily due to issues such as small object recognition, occlusion handling, and stringent real-time processing requirements.

Peppers, an economically vital crop grown and consumed worldwide, are crucial to global agricultural economies. Throughout their cultivation cycle, diseases and pests emerge as the main threats, causing significant annual economic losses for both individual farmers and the broader pepper industry [6]. For instance, diseases

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like powdery mildew and anthracnose can induce premature leaf drop and stunted plant growth, thereby reducing yield and quality while inflating production costs [7]. Consequently, timely and precise detection and identification of pepper diseases and pests are essential—not only for curbing economic losses but also for minimizing pesticide usage and alleviating environmental burdens, which are critical steps toward achieving sustainable agriculture [8].

Currently, pepper diseases and pests detection faces multiple challenges that limit effective quality control. Conventional detection techniques methodologies hinge on manual inspection, which is subject to subjective judgment and experiential differences, often resulting in inconsistent outcomes among different inspectors [9]. Moreover, manual inspection has limited coverage and requires periodic implementation, potentially leading to delays, thereby exacerbating their spread [10]. Additionally, variations in natural conditions, such as variations in weather and lighting conditions, pose challenges for traditional methods to meet diverse detection requirements [11]. Therefore, developing a more accurate, efficient, and easily applicable method for pepper diseases and pests detection is particularly important.

Propelled by significant advancements in computer vision technology, researchers have initiated efforts to employ various computer-assisted techniques for agricultural diseases and pests identification, introducing innovative methodologies for the swift and precise detection of pepper leaf diseases and pests [12]. However, research in complex natural scenarios remains limited. Current studies primarily focus on classifying and identifying agricultural diseases and pests in images with simple backgrounds, with less research on detection in complex backgrounds. Although some advanced network models can achieve mAP values exceeding 90% in simple backgrounds, their generalization ability is limited in complex environments due to factors such as leaf occlusion, uneven lighting, and small lesion areas, with detection performance urgently needing improvement [13].

Within the domain of deep learning, object detection algorithms are predominantly classified into two categories [14, 15]: one-stage detection algorithms that rely on regression techniques and two-stage detection algorithms that utilize candidate boxes. Mahesh and Mathew [9] enhanced disease detection capability in YOLOv3 by employing dilated convolution layers. Yue et al. [16] improved detection accuracy for small sample data through network modifications and bounding box clustering in YOLOv7.

Existing research primarily focuses on leaves severely affected by diseases and pests, typically with large and sparsely distributed lesions that are easy to detect but

often already in middle to late stages, limiting prevention effectiveness. Simultaneously, seasonal changes and environmental impacts result in a scarcity of available diseases and pests images, presenting challenges for research.

To overcome the obstacles in detecting pepper diseases and pests, this study unveils and improves upon the YOLOv10 model, proposing a novel YOLO-Pepper model. This model aims to enhance detection precision and efficiency, particularly in complex environments. The main contributions of this study include: (1) Construction of a dataset containing 8046 high-quality RGB images of pepper disease and pest, featuring complex backgrounds and diverse samples aimed at reflecting real agricultural scenarios; (2) Proposal of the YOLO-Pepper model, which significantly improves pepper disease and pest detection performance in complex environments through the introduction of an Adaptive Multi-Scale Feature Extraction (AMSFE) module, Dynamic Feature Pyramid Network (DFPN), Small Detection Head (SDH), and Inner-CIoU loss function; (3) Validation of the proposed model's effectiveness through extensive experiments, achieving 94.26% mAP@0.5 and 115.26 FPS real-time detection performance while maintaining lightweight characteristics (2.51 M parameters).

Literature review

Object detection methods

Object detection, a pivotal research area in computer vision, encompasses both traditional methodologies and deep learning-based strategies. Traditional techniques, such as combining SIFT and HOG features with SVM for object detection [17], still have limitations in detection accuracy and generalization ability when dealing with complex backgrounds [18–21, 51].

The YOLO algorithm family has achieved significant recognition for its rapid processing capabilities and high precision. Successive epochs, such as YOLOv4 [22], YOLOv5 [23], YOLOX [25], YOLOv6 [26], YOLOv7 [27], YOLOv8 [28], and YOLOv9 [27], have progressively enhanced performance metrics. Network lightweighting has become a key research direction, with recent approaches balancing parameter efficiency and accuracy using hybrid attention mechanisms and fine-grained feature extraction techniques [24]. The most recent version, YOLOv10, incorporates multiple innovative features, significantly boosting both detection accuracy and processing speed [29–31]. However, existing research still shows limitations in handling small objects, complex background interference, and multi-scale targets [32, 33].

Agricultural diseases and pests detection methods

Initially, agricultural diseases and pests detection relied on experts' visual observation of infected plants. This approach was characterized by significant demands on both time and labor, and its effectiveness was heavily reliant on the practitioner's expertise [34]. Subsequently, image recognition methods using manually designed features were adopted, primarily leveraging expert knowledge to manually locate disease areas and determine disease types based on color, texture, and other information [35]. Traditional machine learning methods overcame some inherent misjudgments of manual diagnosis and reduced resource requirements, but they still faced difficulties and couldn't automatically learn and extract features from raw images [36]. Therefore, although these manually designed feature detection methods could identify some diseases and pests, their subjective nature limited their application in real scenarios [37], especially in environments with significant noise such as uneven lighting.

Over the past few years, the academic community has extensively focused on deep learning, owing to its advantages in automatic learning and feature extraction, making significant progress in agricultural diseases and pests recognition [38]. Current research methods have gradually evolved from traditional image processing techniques to deep learning models [39, 40].

During the initial integration of deep learning techniques into agricultural practices, researchers initially used classification models like AlexNet and GoogLeNet to diagnose agricultural diseases and pests in simple backgrounds, continuously optimizing network structures and proposing new deep learning models to pursue stronger feature extraction capabilities and better model performance [41]. Kamal et al. [42] reported a classification accuracy of 98.34% utilizing a Reduced MobileNet architecture trained on the PlantVillage dataset. In a separate study, Wei et al. [43] introduced an enhanced

ResNet model tailored for fruit disease classification, achieving notable results using ResNet-CBAM model incorporating CBAM attention mechanism achieving an accuracy of 99.89%. Although these studies achieved good accuracy in simple background single disease image classification tasks, they may still lack performance in actual complex scenarios.

Detecting agricultural diseases and pests in natural environmental backgrounds is a challenging task. Sun et al. [44] reconstructed common 3×3 convolutions based on the SSD object detection algorithm, achieving 83.12% mAP on a self-built apple disease dataset with complex backgrounds. As concerns about model feasibility in practical applications increase, designing lightweight network models suitable for terminal devices has become a new research trend. Wang et al. [45] introduced the MGA-YOLO, achieving a maximum mAP of 94.0% and reaching 12.5 FPS when tested on mobile phones. In a subsequent advancement, Dong et al. [46] enhanced the original framework through adding effective channel attention, proposing an accurate and robust pest detection model PestLite, which achieved 90.7% mAP. Yue et al. [16] made the first attempt to apply YOLOv7 to detect pepper diseases and pests, achieving outstanding results compared to previous agricultural disease detection algorithms.

Research challenges

Compared to early agricultural diseases and pests detection algorithms, deep learning-based object detection techniques have surpassed the limitations of human experience dependency, not only quickly and accurately locating disease targets but also substantially enhancing accuracy. Nevertheless, they continue to face multiple challenges in practical applications, specifically including: (A) Within-class variation, (B) Cross-class resemblance, (C) Low image quality, and (D) Obstruction (as shown in Fig. 1). For example, changes in camera angles

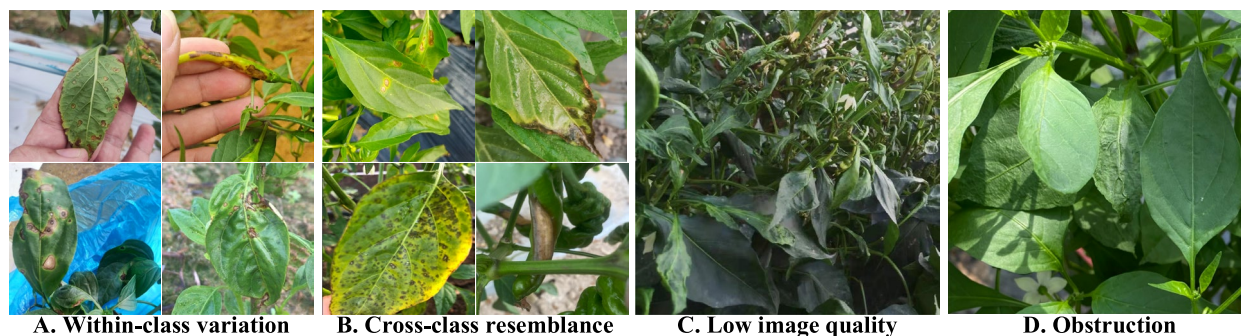


Fig. 1 The challenges of agricultural diseases and pests detection task. **A** Within-class variation; **B** Cross-class resemblance; **C** Low image quality; **D** Obstruction

can cause significant differences in the appearance of the same disease from different perspectives; different diseases may be difficult to distinguish due to similar appearances from similar angles; or images may lack partial information due to perspective obstruction, affecting the comprehensive efficacy of the detection model.

Although there are currently rich datasets on the internet regarding vegetables, fruits, and grain crop diseases and pests, high-quality datasets specifically for pepper diseases and pests remain scarce, particularly with limited research on pepper diseases and pests object detection in complex environments. When detecting pepper diseases and pests in real-world settings, elements like intricate backgrounds, variable lighting conditions, and leaf occlusions further increase detection difficulty. Addressing these challenges, this research introduces a model for detecting pepper diseases and pests, grounded in YOLOv10, introducing multiple innovative modules aimed at improving model detection accuracy and efficiency in complex greenhouse environments.

Materials and methods

Materials

We constructed a specialized pepper disease and pest detection dataset capturing authentic greenhouse cultivation challenges. The dataset comprises 8046 high-resolution (1920×1080) RGB images systematically collected from six commercial greenhouses in Shouguang City, Shandong Province, China spanning the 2022–2023 growing seasons. Data collection was meticulously designed to encompass environmental diversity, with images captured across three distinct growth stages (seedling, flowering, fruiting) and varying weather patterns (sunny, rainy, foggy). This comprehensive environmental coverage ensures the dataset's robust representation of real-world agricultural scenarios.

To maximize sample diversity, we selected surveillance videos from six different greenhouses, each with distinct environmental conditions, planting densities, and management practices. This selection strategy ensures the dataset encompasses various growth stages, disease progressions, and infestation levels commonly encountered in commercial pepper production.

The final dataset contains 8046 high-quality annotated images, extracted using a systematic sampling approach where one frame was saved every 30 frames from each source video as a key frame. All images were manually annotated using Labelling software for precise target detection box generation. This experiment annotated 9 common categories, including 8 disease and pest categories and healthy samples as a control group: healthy, anthracnose, phytophthora blight, viral disease, leaf spot

disease, root rot, blossom-end rot, mites, and thrips, with corresponding examples shown in Fig. 2.

Table 1 provides a detailed breakdown of the dataset composition, enumerating both the quantity of images and annotation boxes corresponding to each category. This distribution reveals the balanced nature of the dataset, with sufficient representation across all classes to support effective model training. To facilitate comprehensive model evaluation, the dataset was systematically partitioned into training, validation, and testing subsets following an 8:1:1 ratio, ensuring that each subset maintained the original class distribution proportions.

The improved pepper diseases and pests detection model based on YOLOv10

YOLOv10 family consists of six variants of increasing size: YOLOv10n, YOLOv10 s, YOLOv10 m, YOLOv10b, YOLOv10 l, and YOLOv10x.

Considering the requirement for efficient and lightweight model deployment on edge devices in agricultural applications, in this study, YOLOv10n was chosen as the base model. YOLOv10 is organized into three core components: the backbone network, the neck network, and the detection head (Fig. 3). Figure 4 illustrates the complete network structure of YOLOv10n, including three main components: Backbone, Neck, and Head, as well as the hierarchical structure and connection methods within each component.

The task of detecting pepper diseases and pests in complex greenhouse environments requires simultaneous detection of different types of diseases and pests targets. Each diseases and pests target has variable scale, often with small sizes. Additionally, the numerous objects in pepper greenhouses, along with different types and occlusion overlaps, make it difficult to estimate their impact on detection performance. These challenges need to be addressed specifically. Therefore, the model requires lightweight improvements to adapt to resource-constrained devices. In response to these issues, this research introduces an enhanced YOLOv10 algorithm, named YOLO-Pepper (Fig. 4).

Figure 4 illustrates that the YOLO-Pepper model markedly improves detection performance while minimizing computational demands. This is achieved through the optimization of the backbone network, feature fusion module, and detection head within the YOLOv10n framework. These enhancements bolster the model's suitability for deployment on edge devices in greenhouse settings.

Adaptive multi-scale feature extraction (AMSFE) module

The AMSFE structure draws inspiration from Lin et al.'s [47] Multi-Head Mixed Convolution (MHMC)



Fig. 2 Representative samples from each category in the pepper diseases and pests dataset

architecture. This design builds upon GhostNet's principle [48] that redundancy-rich feature maps contain valuable information that can be efficiently extracted using parameter-efficient structures. GhostNet divides input feature maps into two parts—a primary path and cheap operations—to reduce computational complexity. RepConv [49] is a model reparameterization technique that consolidates branch parameters into the main branch

during inference, enhancing model efficiency and performance across various tasks, including object detection. The AMSFE structure is illustrated in Fig. 5.

Figure 5 demonstrates that following the passage of input features through RepConv for branch feature fusion, AMSFE segments the features into four groups based on channel number. Each group is subjected to convolution operations with varying kernel sizes, and

Table 1 Sample distribution in the pepper diseases and pests dataset

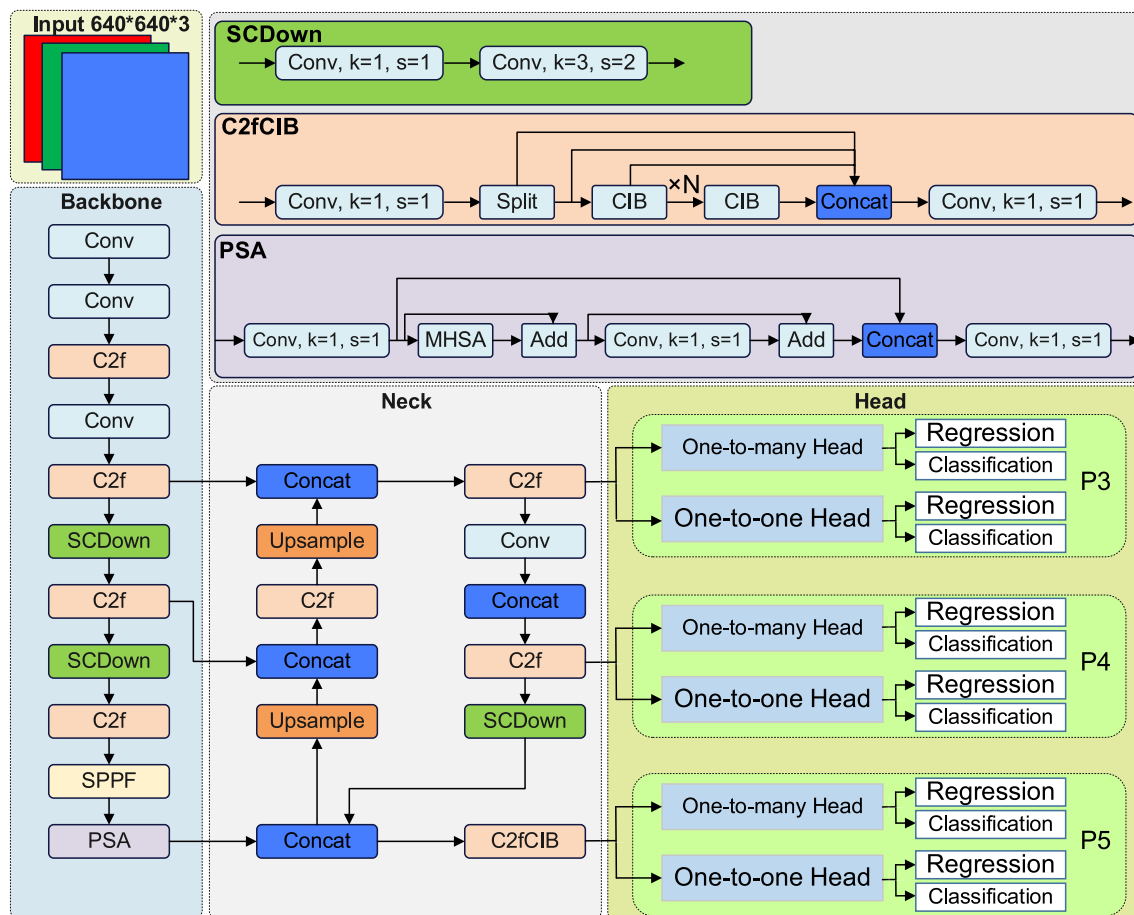
Class	Number of Images	Number of Annotations
Healthy	1328	2744
Anthraxnose	856	1740
Blight	768	1583
Virus	779	1651
Spot	664	1357
Root Rot	897	1818
Fruit Rot	930	1917
Mites	968	1973
Thrips	856	1716
Total	8046	16,499

the resultant outputs are concatenated to generate new feature maps. Group convolution allows each subset to perform independent convolution operations, leading to feature isolation between different subgroups. Therefore,

Pointwise Conv [50] is employed after group operations to learn cross-channel correlations and perform weighted fusion of features from different channels. The AMSFE structure enhances feature extraction and gradient flow through reparameterized convolution in the gradient flow branch while utilizing multi-scale convolutions internally. Through parallel processing, the structure maintains a relatively low parameter count, making it suitable for edge device deployment. Considering that deeper feature maps better leverage the advantages of the AMSFE structure, only the last three layers of C2f structures in the backbone are replaced.

Dynamic feature pyramid network (DFPN)

The initial YOLOv10 architecture employs the PANet structure [19, 51] within the Neck segment to facilitate feature fusion. Traditional FPN structures adopt a top-down approach, only considering feature map fusion between adjacent levels and scales. While the PANet structure addresses some limitations of traditional FPN, semantic information can still be easily lost during

**Fig. 3** Architecture of the baseline YOLOv10n model

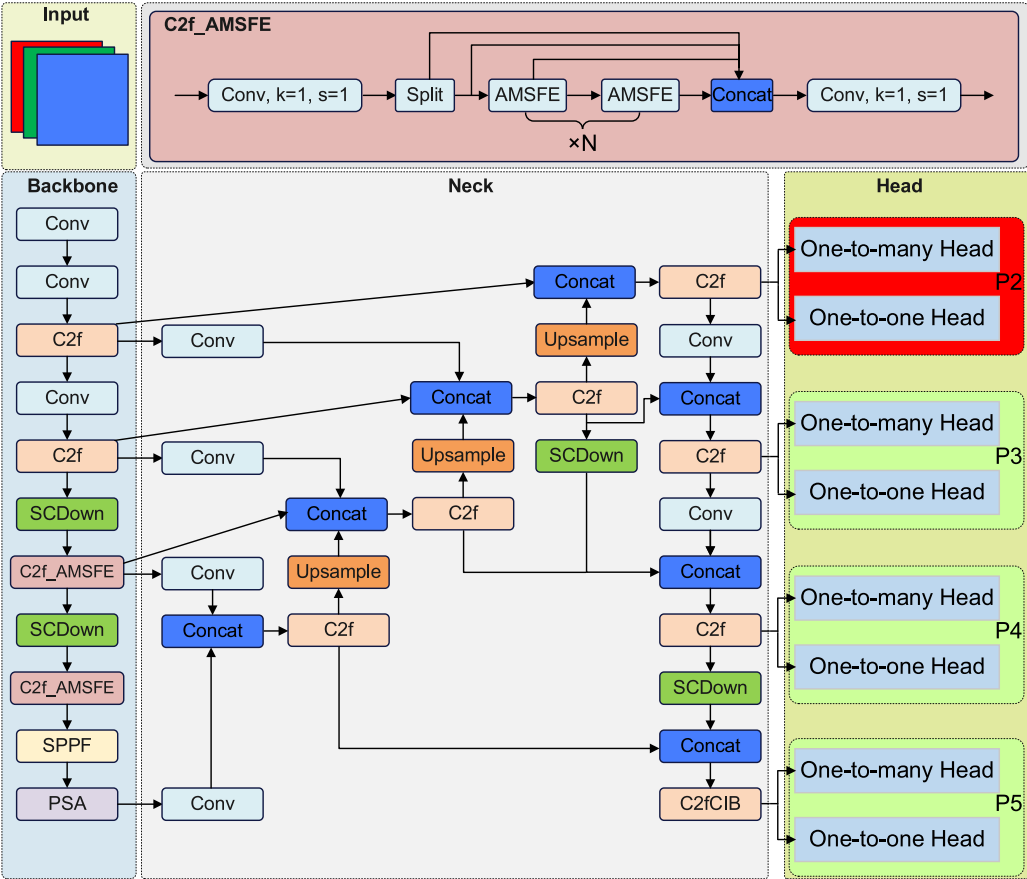


Fig. 4 Architecture of the proposed YOLO-Pepper model

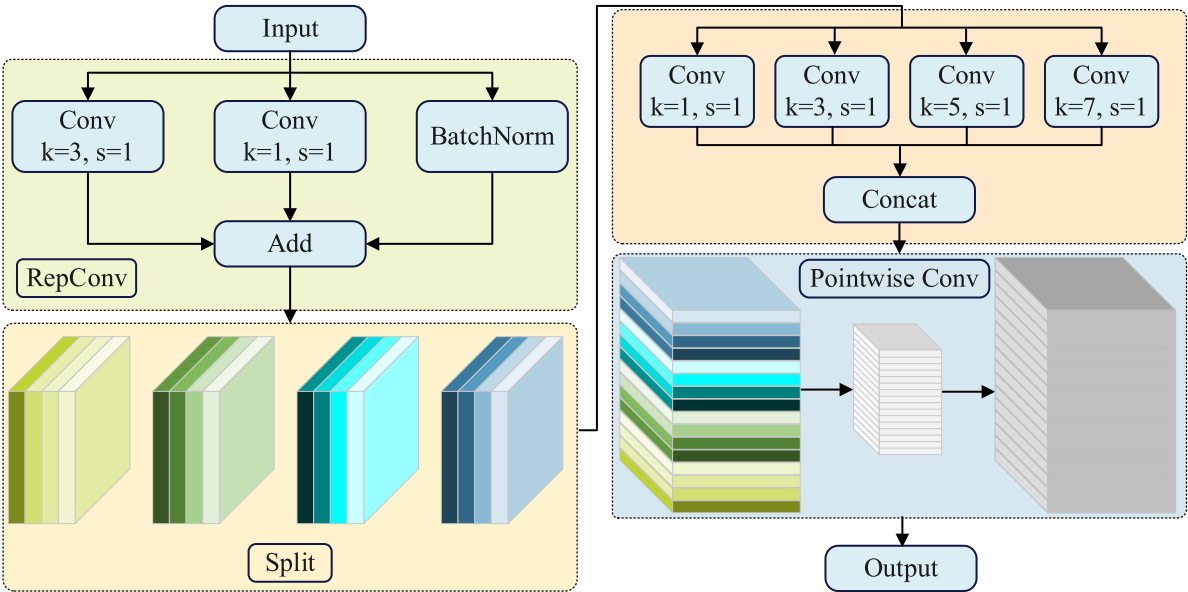


Fig. 5 Structure of the adaptive multi-scale feature extraction (AMSFE) module

downsampling and upsampling processes when detecting small objects. This study introduces the DFPN structure. This structure integrates various levels and scales via cross-scale connections, enabling the concurrent processing of both high-level and low-level semantic information. Figure 6 illustrates the structural comparisons between FPN, PANet, and DFPN.

Figure 6 illustrates that the DFPN architecture provides low computational complexity and decreased inference time, efficiently handling both high-level semantic information and low-level spatial information with equal importance, achieving feature fusion across different depths. In Fig. 7, dotted lines represent cross-layer feature matrix transmission. The cross-layer feature matrix transmission process in DFPN is expressed by Eq. (1), where k represents the pyramid structure level, and P_k^i represents the feature at the i -th position within the k -th layer.

$$P_k^i = \text{Conv}(\text{Concat}(P_1^i, P_2^i, \dots, P_{k-1}^i)) \quad (1)$$

In Eq. (1), *Concat* represents feature concatenation, and *Conv* denotes convolution operation.

The DFPN structure's specific process for cross-layer concatenation of large-scale and small-scale feature maps is shown in Eq. (2). Channel matching is achieved through downsampling of large-scale features via max pooling and upsampling of small-scale features through linear interpolation.

$$P_{k+1}^i = \text{Conv}(\text{Concat}(\text{MaxPool}(P_{k-1}^i), P_k^i, \text{Bilinear}(P_k^{i+1}))) \quad (2)$$

where *MaxPool* represents the max pooling operation, and *Bilinear* denotes bilinear interpolation upsampling.

Small-scale detection head (SDH)

The original YOLOv10 model incorporates three detection scales: 80×80 , 40×40 , and 20×20 , corresponding to layers P3, P4, and P5, respectively. These detection layers with different sizes have varying receptive fields for detecting objects of different scales. In greenhouse

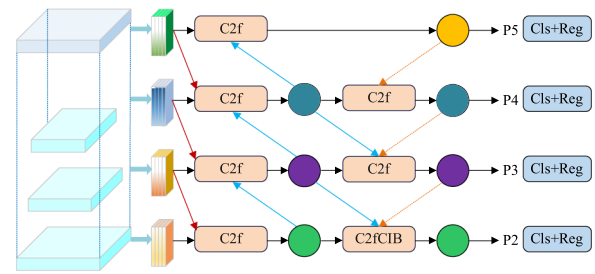


Fig. 7 DFPN-P2P3P4P5 structure

cultivation environments, pepper diseases and pests targets constitute a minor fraction of the overall image area, with most being small objects. While small objects are numerous in the images, feature maps generated by large object detection layers contain limited feature information for small targets due to high downsampling ratios, resulting in poor detection performance. Addressing these issues and considering the limited effectiveness of large object detection heads in small object recognition tasks, this study proposes adding a P2 layer dedicated to small object detection, aimed at improving the accuracy of detecting small objects while maintaining model efficiency. This results in the DFPN-P2P3P4P5 structure shown in Fig. 7.

As illustrated in Fig. 7, the improved model's Neck section employs the proposed DFPN structure with an additional P2 small object detection layer. Downward-slanting arcs represent upsampling operations, while upward-slanting lines indicate downsampling operations.

Inner-CIoU

The loss function of YOLOv10 is composed of three distinct elements:

$$\text{Loss} = \text{Loss}_{\text{box}} + \text{Loss}_{\text{dfl}} + \text{Loss}_{\text{cls}} \quad (3)$$

where Loss_{box} is the regression loss for predicted boxes, Loss_{dfl} is the position loss for offsets, and Loss_{cls} is the classification loss. To quantify the positional discrepancies between the predicted and ground truth bounding

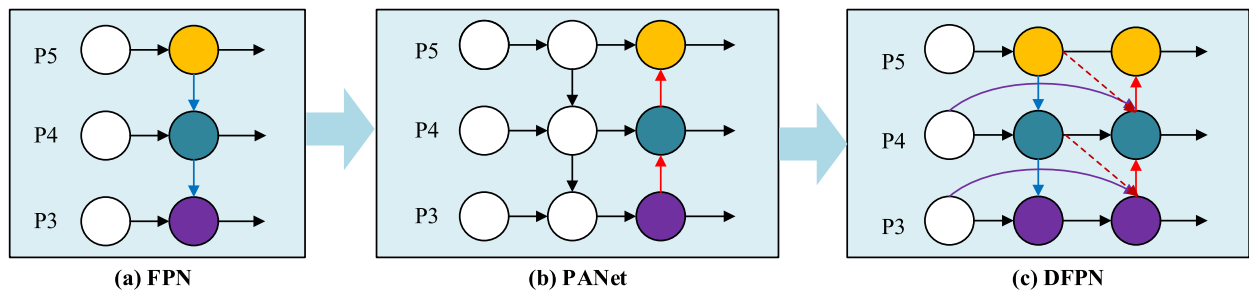


Fig. 6 Comparison of feature fusion structures

boxes, YOLOv10n employs CIoU loss function, defined as:

$$Loss_{CIoU} = 1 - IoU + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v \quad (4)$$

$$\alpha = \frac{v}{(1 - IoU) + v} \quad (5)$$

$$v = \frac{4}{\pi^2} \left(\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h} \right)^2 \quad (6)$$

Here, $\rho^2(b, b^{gt})$ signifies the Euclidean distance between the centers of two points, α serves as a weighting factor, v indicates the aspect ratio similarity between the predicted and ground truth boxes, c denotes the diagonal length of the smallest enclosing box, w^{gt} and h^{gt} represent the width and height of the ground truth box, respectively, while w and h correspond to the width and height of the predicted box.

The standard CIoU loss function, while effective for regular objects, shows limitations when applied to pepper diseases and pests that exhibit variable scales and irregular shapes. Specifically, when calculating Intersection over Union (IoU), CIoU's consideration of center points and aspect ratios fails to adequately balance localization accuracy across different scales, compromising detection precision.

As shown in Fig. 8, auxiliary bounding boxes share the same center as actual bounding boxes but differ in size. Variations in the Intersection over Union (IoU) metrics of auxiliary bounding boxes are indicative of corresponding fluctuations in the IoU metrics of the primary bounding

boxes. In instances of high IoU, the absolute gradient values of the IoU for smaller auxiliary bounding boxes surpass those of the primary bounding boxes. Conversely, for low IoU instances, the absolute gradient values of the IoU for larger auxiliary bounding boxes are inferior to those of the primary bounding boxes, accelerating convergence through auxiliary bounding box size variations. The auxiliary bounding boxes are adjusted using a ratio factor ranging from [0.5, 1.5]. This ratio allows for auxiliary bounding box scale adjustment according to different datasets, enhancing model generalization capability. Through auxiliary box scale transformation, the model can focus more on fine-tuning prediction box positions when handling high IoU pepper diseases and pests targets, thereby improving localization accuracy. When processing low IoU targets, using larger-scale auxiliary boxes effectively covers target regions, reducing localization bias caused by inappropriate scaling.

Applying Inner-IoU [52] to the existing CIoU, Inner-CIoU is introduced, as shown in Eqs. (7) through (18).

Equations (7) through (10) calculate the respective coordinates for the left, right, top, and bottom boundaries of the actual auxiliary bounding box.

$$b_t^{gt} = y_c^{gt} - \frac{h^{gt} \cdot ratio}{2} \quad (7)$$

$$b_b^{gt} = y_c^{gt} + \frac{h^{gt} \cdot ratio}{2} \quad (8)$$

$$b_l^{gt} = x_c^{gt} - \frac{w^{gt} \cdot ratio}{2} \quad (9)$$

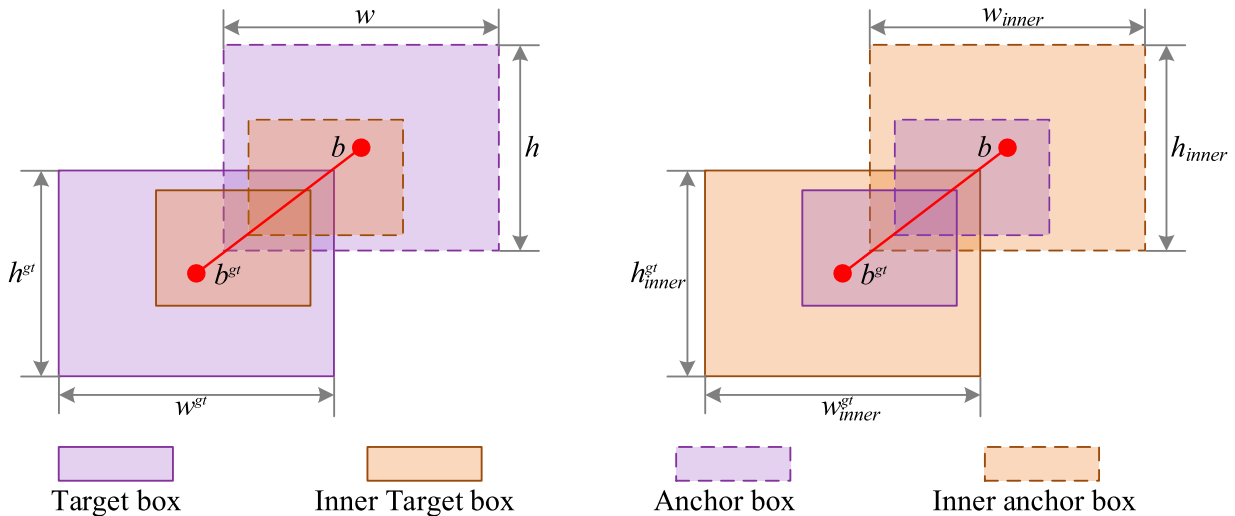


Fig. 8 Inner-CIoU loss function

$$b_r^{gt} = x_c^{gt} + \frac{w^{gt} \cdot ratio}{2} \quad (10)$$

Equations (11) through (14) calculate the coordinates for the left, right, top, and bottom boundaries of the predicted auxiliary bounding box, respectively.

$$b_l = x_c - \frac{w \cdot ratio}{2} \quad (11)$$

$$b_r = x_c + \frac{w \cdot ratio}{2} \quad (12)$$

$$b_t = y_c - \frac{h \cdot ratio}{2} \quad (13)$$

$$b_b = y_c + \frac{h \cdot ratio}{2} \quad (14)$$

Equations (15) and (16) calculate the intersection area and union area of auxiliary boxes, respectively.

$$inter = \left(\min(b_r^{gt}, b_r) - \max(b_l^{gt}, b_l) \right) \cdot \left(\min(b_b^{gt}, b_b) - \max(b_t^{gt}, b_t) \right) \quad (15)$$

$$union = (w^{gt} \cdot h^{gt}) \cdot (ratio)^2 + (w \cdot h) \cdot (ratio)^2 - inter \quad (16)$$

Equations (17) and (18) calculate the IoU of auxiliary boxes and the total Inner-CIoU loss function, respectively.

$$IoU^{inner} = \frac{inter}{union} \quad (17)$$

$$Loss_{Inner-CIoU} = Loss_{CIoU} + IoU - IoU^{inner} \quad (18)$$

Considering the multi-scale nature of pepper diseases and pests targets, this study set the ratio scale factor to 1.2 after testing, which better focuses the auxiliary bounding box shape on the inherent scale of pepper diseases and pests targets. This adjustment enables better adaptation to various shapes of pepper diseases and pests in greenhouse environments.

Results and discussion

Experimental environment

All experimental procedures were executed on a system running Windows 11, outfitted with an Nvidia GeForce 1080 Ti graphics card. The model framework utilized PyTorch 2.2.2, Python 3.9, and CUDA 11.8.

The input images were resized to 640 × 640 pixels, processed in batches of 16 over 200 epoches. The initial learning rate was configured at 0.001. To mitigate overfitting, a weight decay strategy was applied to the network parameters, with a decay coefficient of 0.0005. Optimization for bias and BatchNorm layer weights was conducted using Stochastic Gradient Descent with Momentum (SGD + Momentum).

Evaluating indicator

To evaluate the performance of various models in detecting diseases and pests, standard metrics were utilized, including Precision, Recall, F1-score (harmonic mean), and mean Average Precision (mAP). Specifically, mAP_{0.5}, which denotes the mAP at an IoU threshold of 0.5, was employed. Model complexity was assessed through metrics such as the size of the weight files (Weights), the number of parameters (Params), and the Frames per Second (FPS).

Learning rate selection

The learning rate is a pivotal hyperparameter that profoundly influences the convergence and predictive capability. An excessively high learning rate can lead to slow iteration and failure to converge, while an excessively low rate may cause overfitting. An appropriate learning rate facilitates rapid model convergence, making its selection crucial for performance optimization. Based on experimental research and experience, learning rates of 0.1, 0.05, 0.01, and 0.001 were evaluated. The loss and mAP values under different learning rates are illustrated in Fig. 9.

Figure 10 illustrates that, given an identical number of epoches, the model's prediction loss and accuracy exhibited negligible differences between learning rates of 0.1 and 0.05. Conversely, with a learning rate of 0.001, the loss decreased more swiftly during training, enabling the model to converge more rapidly and attain relatively optimal mAP predictions. Consequently, a learning rate of 0.001 was chosen for model construction and subsequent parameter optimization.

Model training

To assess the efficacy of the proposed enhanced model, evaluations were performed using a bespoke dataset of pepper diseases and pests. Figure 10 depicts the convergence trajectories of several performance metrics. The x-axis represents the number of epoches. As shown in Fig. 10, after 80 epoches, all loss curves approached zero, indicating model convergence. Ultimately, the box_loss value stabilized at approximately 0.034.

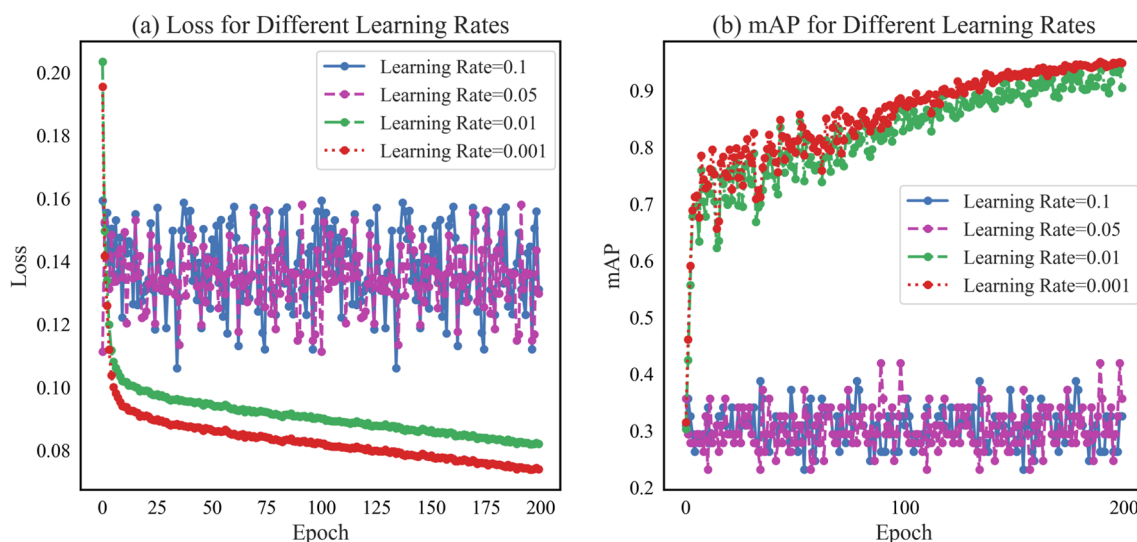


Fig. 9 Loss values and mAP of YOLO-pepper model under different learning rates

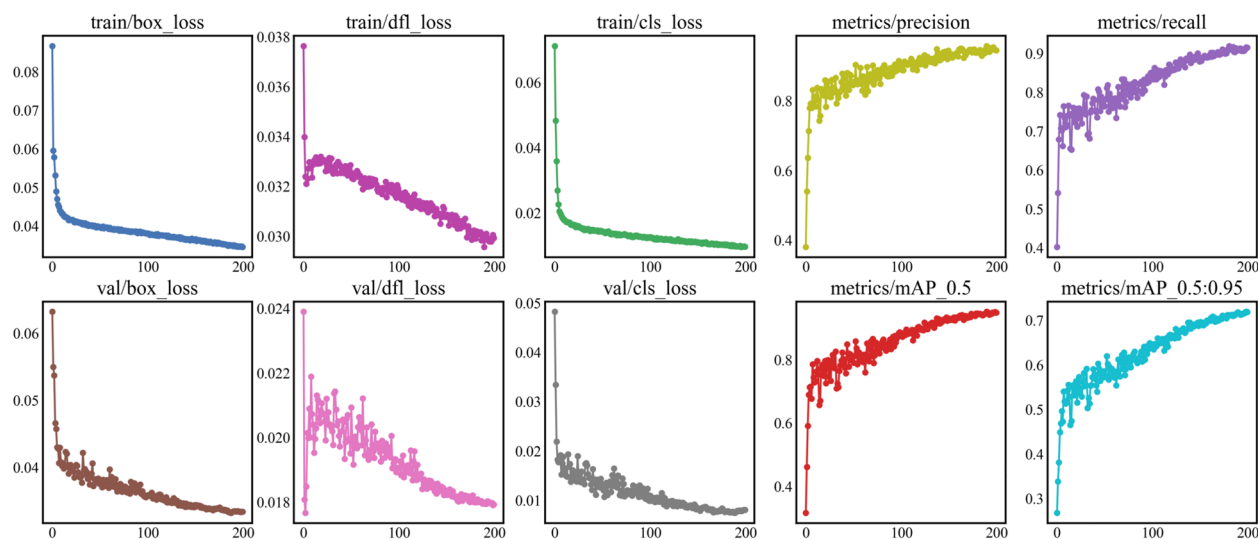


Fig. 10 Performance indicator curves of the model in the training and validation sets

Analysis of the convergence curves reveals that after 100 epoches, the curves entered convergence, with mAP@0.5 values stabilizing around 94% and mAP@0.5:0.95 values stabilizing around 70%.

Ablation study results

To thoroughly assess the efficacy of each enhancement strategy within the YOLO-Pepper model, a series of meticulously designed ablation experiments were conducted, with the outcomes detailed in Table 2. The initial row outlines the YOLOv10n model as the baseline, showcasing the detection performance and primary metrics of the original framework. Experiment A augmented

the final three C2f structures in the Backbone section by integrating the C2f_AMSFE structure to bolster feature extraction capabilities. Experiment B introduced the DFPN structure within the Neck section. Experiment C incorporated an additional detection head specifically tailored for small object detection. Experiment D further refined the model's learning process by substituting the CIoU bounding box loss function with Inner-CIoU.

The ablation study results in Table 2 clearly demonstrate each improvement strategy's impact. Experiment A shows that integrating the C2f_AMSFE structure boosted mAP by 2.18 percentage points to 84.56%. This significant accuracy improvement came at minimal

Table 2 Ablation experiment results

Models	mAP@0.5/%	Params/M	FPS	Weights/M
Baseline	82.38	2.72	109.11	5.75
A	84.56	2.83	103.32	5.78
A + B	89.75	2.43	123.36	4.92
A + B + C	92.18	2.51	115.28	5.15
A + B + C + D	94.26	2.51	115.26	5.15

computational cost—only a 4% increase in parameter count (from 2.72 M to 2.83 M)—highlighting the C2f_AMSFE structure's efficiency.

The results of Experiment A + B are particularly noteworthy. After incorporating the DFPN structure in the Neck section, the model's performance exhibited a qualitative leap: the mAP value increased substantially by 5.19 percentage points, reaching a high accuracy of 89.75%. Even more impressively, this improvement not only enhanced detection accuracy but also achieved significant model lightweighting. Specifically, the parameter count dropped dramatically from 2.83 M to 2.43 M, a reduction of 14.13%; model weights decreased from 5.78 M to 4.92 M, a reduction of 14.88%. Simultaneously, detection speed increased significantly, with FPS rising from 103.32 to 123.36, an improvement of 19.39%. These results strongly demonstrate the exceptional efficacy of the DFPN structure in object detection tasks, perfectly achieving multiple objectives of improved accuracy, model lightweighting, and speed optimization.

The results of Experiment A + B + C validate the efficacy of the small detection head strategy. The mAP value again increased significantly by 2.43 percentage points, reaching 92.18%. Although the model's parameter count increased slightly (from 2.43 M to 2.51 M) to support this new functionality. This compromise is fully warranted given the significant enhancement in performance. Remarkably, even with the integration of an additional detection head, the model sustained a high frame rate of 115.28 FPS.

Experiment A + B + C + D results showcase the impact of the Inner-CIoU. The mAP value saw an additional increase of 2.08 percentage points, culminating in an outstanding performance of 94.26%. Notably, this enhancement in accuracy was achieved without any increase in parameters or model weight, while preserving a high detection speed exceeding 115 FPS. This clearly illustrates the dual benefits of the Inner-CIoU loss function in enhancing detection accuracy and maintaining computational efficiency.

In conclusion, YOLO-Pepper achieved significant progress across multiple key indicators, providing a solution

Table 3 Comparative experimental results

Methods	F1-score	mAP@0.5/%	Params/M	FPS	Weights/M
SSD	58.76	65.92	26.83	85.62	31.45
FasterR-CNN	65.92	71.84	141.56	19.73	93.82
YOLOv3	72.45	77.89	61.83	43.26	121.54
YOLOv5 s	73.88	78.95	7.23	108.45	14.86
YOLOX-s	76.54	80.67	8.92	102.34	15.73
YOLOv7-tiny	77.83	81.92	6.85	113.62	10.24
YOLOv8n	79.56	83.78	4.32	116.85	8.76
YOLOv9-tiny	80.93	84.89	3.15	121.43	6.35
YOLOv10n	80.56	82.38	2.72	109.11	5.75
YOLO-Pepper	91.76	94.26	2.51	115.26	5.15

for pepper diseases and pests detection that combines high accuracy, efficiency, and practicality.

Comparative experiments

To validate the efficacy of YOLO-Pepper in identifying pepper diseases and pests within greenhouse settings, we performed extensive comparisons against the latest YOLOv10n and other prominent object detection algorithms under uniform experimental conditions and datasets. Table 3 delineates the experimental outcomes, vividly showcasing the evolutionary traits and performance disparities among the various detectors.

The data presented in Table 3 indicate that YOLO-Pepper markedly surpasses other models in various performance metrics for detecting pepper diseases and pests within greenhouse environments. Regarding detection efficacy, conventional two-stage detectors (such as Faster R-CNN) and single-stage detectors (such as SSD) show relatively weak performance, with F1-scores of only 65.92% and 58.76%, respectively, and mAP@0.5 reaching only 71.84% and 65.92%. This limitation primarily stems from these early models' constraints in handling complex scenes and small objects. Notably, while Faster R-CNN's two-stage design improves detection accuracy, its substantial parameter count of 141.56 M and low FPS of 19.73 severely restrict its practical application.

The evolution of the YOLO series reflects significant trends in object detection development, from YOLOv3 to the latest YOLOv10n, and finally to our proposed YOLO-Pepper. These models continuously seek the ideal equilibrium between precision and computational efficiency, specifically manifested in:

- (1) Continuous parameter reduction: Parameters decreased from 61.83 M in YOLOv3 to 2.51 M in

YOLO-Pepper, a 95.9% reduction. YOLO-Pepper's parameter count is 7.7% lower than YOLOv10n and only 1.77% of Faster R-CNN's. The weight file size, at 5.15 M, is the smallest among all compared models.

- (2) Significant speed improvement: FPS increased from 43.26 to 115.26, a 166.4% improvement. YOLO-Pepper's FPS is 5.6% higher than YOLOv10n and 5.84 times that of Faster R-CNN, only slightly lower than YOLOv9-tiny's 121.43 FPS.
- (3) Steady accuracy enhancement: F1-score improved from 72.45% to 91.76%, and mAP@0.5 increased from 77.89% to 94.26%. Compared to YOLOv10n, YOLO-Pepper achieved improvements of 11.20 and 11.88 percentage points in F1-score and mAP@0.5, respectively, demonstrating its exceptional performance in pepper diseases and pests detection.

Further experiments validate YOLO-Pepper's performance through focused mAP and Loss curve comparisons with the baseline YOLOv10n model (Fig. 11). Figure 11a illustrates the clear superiority of YOLO-Pepper's mAP curve, which shows more rapid convergence and ultimately stabilizes at 94.26%, representing an 11.88 percentage point improvement over YOLOv10n's 82.38%. Figure 11b demonstrates YOLO-Pepper's consistently lower loss values throughout the training process, with an initial loss of 0.197 rapidly decreasing and converging to approximately 0.074 (versus YOLOv10n's 0.083). This comparative analysis reveals both YOLO-Pepper's faster convergence rate

and superior stability, critical factors for efficient model training and robust performance.

YOLO-Pepper exhibits exceptional performance regarding accuracy, speed, and lightweight design, emphasizing its substantial ability for pepper diseases and pests detection in greenhouse environments. The model's low parameter count, small weight file size, and high detection speed further reinforce its practical value in real-world scenarios.

Detection results for various diseases and pests class

To comprehensively evaluate YOLO-Pepper's performance in detecting various pepper diseases and pests, we conducted tests on our imbalanced dataset. Table 4 details the detection outcomes for various diseases and pests types utilizing the YOLO-Pepper model.

Table 4 details the performance metrics of the YOLO-Pepper model in detecting various categories of pepper diseases and pests. Precision represents the proportion of targets that actually belong to a certain category among all targets detected as that category, reflecting the model's accuracy; Recall represents the proportion of correctly detected targets among all targets that actually belong to a certain category, measuring the model's completeness; AP50 (Average Precision at IoU threshold 0.5) comprehensively considers both precision and recall, serving as a key indicator for evaluating the overall performance of object detection models. These metrics are crucial for pepper diseases and pests detection because, in practical applications, there is a need to accurately identify the type of diseases and pests (high precision) to

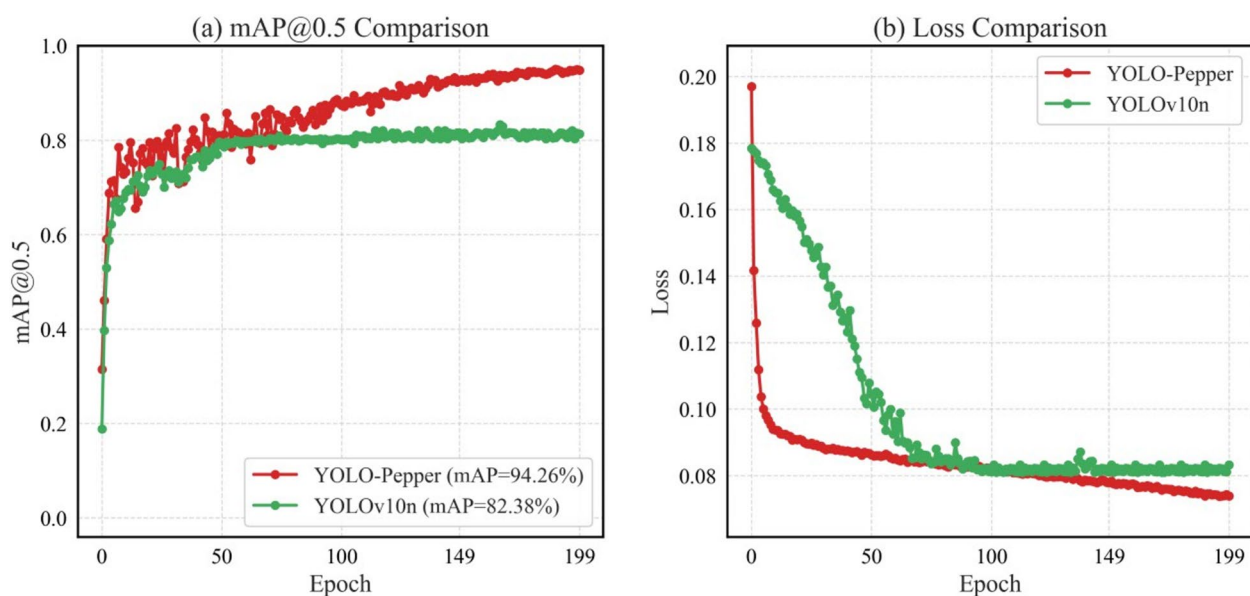


Fig. 11 Comparative analysis of mAP and loss curves across various models

Table 4 Detection results of YOLO-pepper model for various diseases and pests class

Class	Precision (%)	Recall (%)	AP50 (%)
Healthy	98.14	95.92	97.67
Anthrachnose	97.83	94.35	96.00
Blight	94.67	93.45	94.28
Virus	93.78	92.31	93.20
Spot	96.36	94.64	96.01
Root Rot	94.29	93.12	94.23
Fruit Rot	94.56	93.47	94.44
Mites	92.11	90.89	91.69
Thrips	91.72	90.43	90.83
All classes	94.69	93.87	94.26

avoid misdiagnosis and unnecessary treatment, as well as to detect as many diseases and pests targets as possible (high recall) to prevent disease spread due to missed detections.

As shown in Table 4, YOLO-Pepper achieves Precision, Recall, and AP values exceeding 90% for all nine disease/pest categories and healthy samples, demonstrating high precision and recall rates. Notably, the AP reaches 96.01% and 96.00% for Spot and Anthracnose detection, respectively. Comparing detection results across different categories reveals slight variations in detection accuracy. Spot disease exhibits the highest detection precision, while Thrips shows the lowest, primarily due to differences in visual distinguishability of disease features. The results consistently demonstrate YOLO-Pepper's strong learning capability for detecting in greenhouse environments. The overall mAP of 94.26% convincingly proves

its excellent performance in handling different types of pepper diseases and pests.

Figure 12 illustrates the Precision-Recall (PR) curves for YOLO-Pepper and YOLOv10n in detecting nine categories of pepper diseases/pests and healthy samples.

As evident in Fig. 12, while the PR curves for healthy sample detection show relatively small differences between YOLO-Pepper and YOLOv10n, significant variations are observed in the PR curves for the other eight categories of pepper disease and pest. YOLO-Pepper consistently achieves optimal positions across all category PR curves, validating its superior detection capabilities for multi-scale, multi-category pepper disease and pest.

Visualization of feature activation maps and model interpretability

To provide insight into how YOLO-Pepper identifies diseases and pests, we visualized feature activation maps using Grad-CAM to highlight regions the model focuses on during detection. Figure 13 presents these visualizations for representative samples across different disease and pest categories.

Figure 13 reveals that YOLO-Pepper effectively focuses on pathologically relevant regions when detecting different conditions. These visualizations demonstrate that YOLO-Pepper's detection process aligns with the diagnostic criteria used by human experts, focusing on the most relevant visual features for each disease or pest category. Furthermore, comparing activation maps between YOLO-Pepper and YOLOv10n reveals that our enhanced model exhibits more precise attention to symptomatic regions, particularly for small and early-stage symptoms.

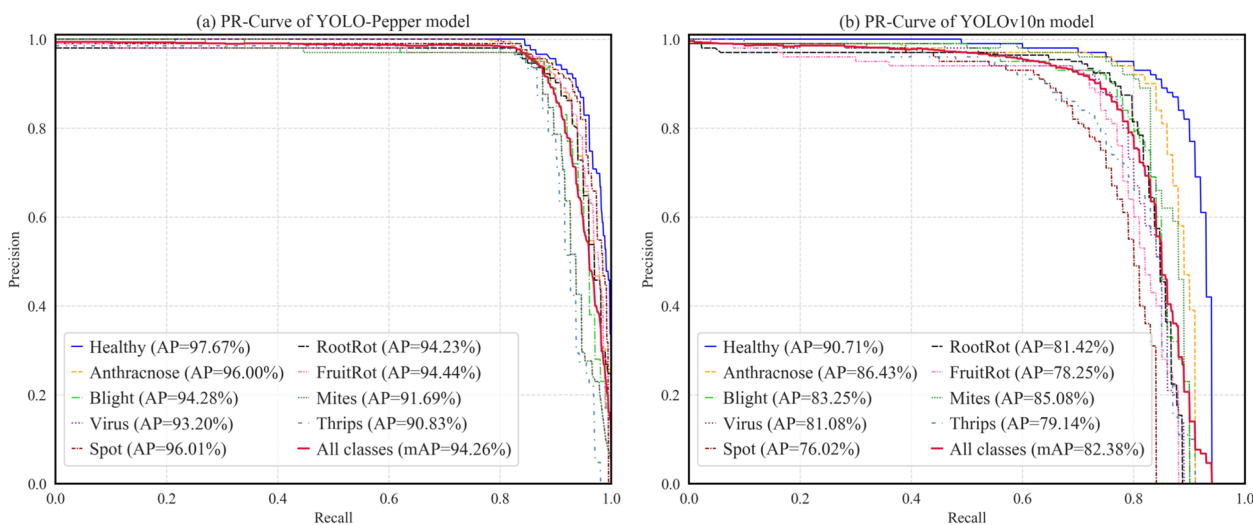


Fig. 12 Precision-recall (PR) curves for different pepper disease and pest detection tasks

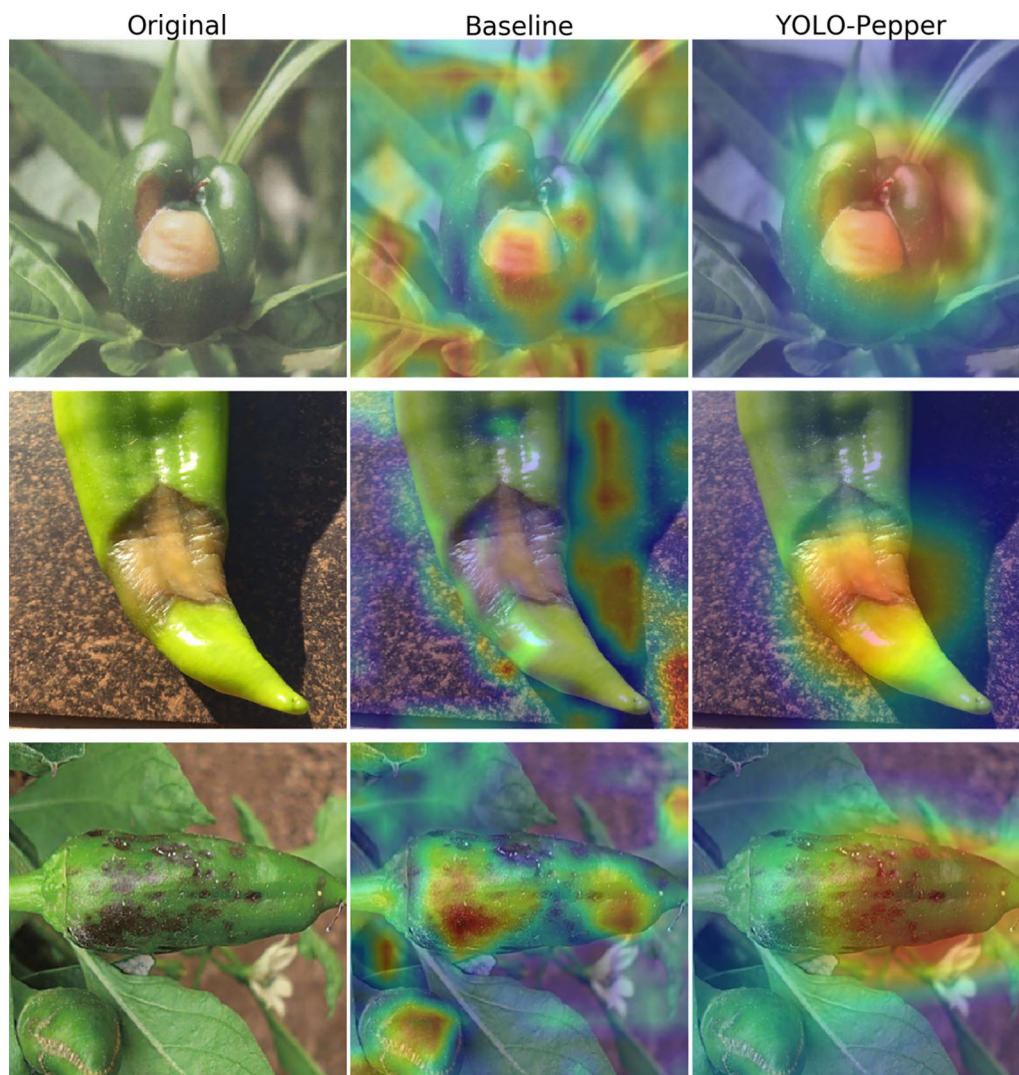


Fig. 13 Comparison of feature activation maps between YOLO-pepper and YOLOv10n

The improved feature focus can be attributed to the AMSFE module's enhanced feature extraction capabilities and the additional P2 detection head for small targets. These visualization results provide interpretable evidence for why YOLO-Pepper achieves superior detection performance, especially for subtle symptoms and complex cases.

Detection performance in real greenhouse environments

To evaluate YOLO-Pepper's practical performance in actual greenhouse conditions, we conducted tests using real pepper plant images exhibiting various diseases and pests. Figure 14 demonstrates the model's detection capabilities across common greenhouse scenarios.

As illustrated in Fig. 14, YOLO-Pepper successfully identifies multiple disease and pest instances in complex greenhouse environments. The top row shows

ground truth annotations of the target conditions, while the bottom row displays our model's detection results with confidence scores. These results confirm YOLO-Pepper's robust performance in real greenhouse environments. The model maintains high detection accuracy for both localized infections and distributed infestations, demonstrating its practical utility for greenhouse management applications.

When compared with the baseline YOLOv10n model (results shown in Table 5), YOLO-Pepper consistently achieves significantly higher detection accuracy across all challenging conditions. This performance advantage is particularly pronounced for early-stage symptoms and cases with high occlusion, where traditional models typically struggle. The results validate YOLO-Pepper's suitability for deployment in greenhouse operations,

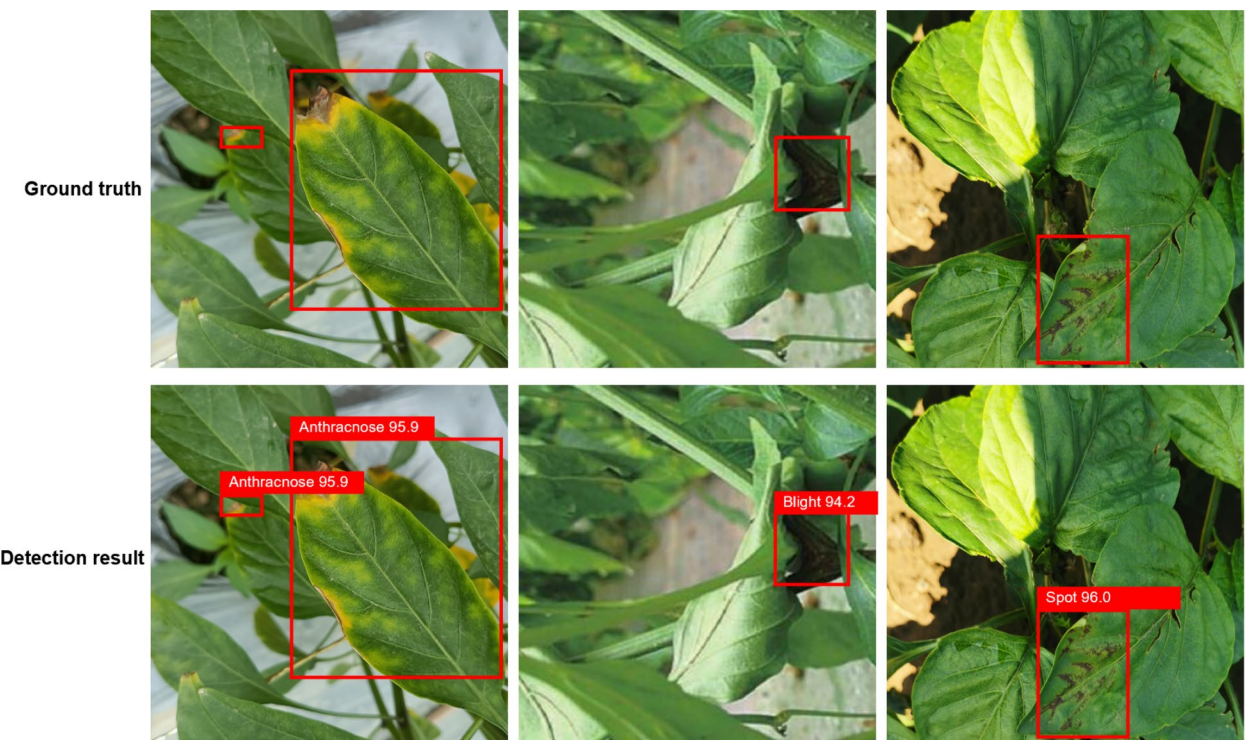


Fig. 14 Detection results of YOLO-pepper in real greenhouse environments

Table 5 Performance comparison in realistic greenhouse conditions (mAP@0.5)

Challenge scenario	YOLO-pepper	YOLOv10n (Baseline)	Improvement (Δ)
Leaf occlusion	92.1	73.8	+ 18.3
Low light	89.4	68.5	+ 20.9
Dense pest clusters	88.7	71.2	+ 17.5
Early-stage symptoms	87.3	64.9	+ 22.4

where early and accurate detection is critical for effective pepper diseases and pests management.

Conclusions and future directions

Conclusion

This study introduces YOLO-Pepper, a specialized detection model designed specifically to overcome the persistent challenges in greenhouse pepper disease and pest monitoring. Our approach directly addresses critical limitations in current detection systems—including plant occlusion, target density, and symptom ambiguity—that typically lead to missed detections, false positives, and reduced accuracy in commercial greenhouse environments. The model incorporates several key improvements: modifying the last three C2f structures in the Backbone to C2f_AMSFE structures, utilizing DFPN structure in the Neck, adding a P2 detection head for

small objects, and implementing Inner-CIoU loss function with auxiliary bounding boxes to effectively enhance small object regression and improve overall detection performance.

Experiments conducted on our custom pepper diseases and pests dataset demonstrate that YOLO-Pepper outperforms baseline and other mainstream models. The model attains an mAP@0.5 of 94.26%, utilizing merely 2.51 million parameters, with a model size of 5.15 MB, and a detection speed of 115.26 FPS. This facilitates swift and precise detection of pepper diseases and pests in greenhouse environments, facilitating timely and precise diagnosis, reducing the risk of large-scale outbreaks, and advancing the application of computer vision technology in smart plant protection.

Future directions

Three primary directions for future research are proposed:

- (1) Further optimization of image preprocessing to account for real greenhouse conditions such as lighting and shadows, incorporating image enhancement and exposure adjustment techniques to improve algorithm robustness in complex lighting environments.

- (2) Additional model lightweighting through techniques such as parameter pruning and knowledge distillation, continually simplifying model structure to achieve further size reduction while maintaining accuracy.
- (3) Ongoing dataset expansion to include diseases and pests data collected under more challenging conditions like nighttime, thereby enhancing the adaptability of diseases and pests detection.

Declaration of generative AI and AI-assisted technologies

Throughout the preparation of this manuscript, the authors utilized various AI tools to enhance language clarity and readability. Subsequently, the authors meticulously reviewed and edited the content as necessary, assuming full responsibility for the final publication.

Author contributions

The research was designed by JL and XW. JL and XW carried out experiments, analyzed data, and drafted the manuscript. The manuscript was revised by XW, QC and JL. All authors reviewed and approved the final version of the manuscript.

Funding

This research was funded by the Shandong Province Natural Science Foundation (Grant Nos. ZR2023MF048, ZR2023QC116 & ZR2021QC173), the Key R&D Program of Shandong Province, China (Grant No. 2024RZB0206), the Disciplinary Construction Funds of Weifang University of Science and Technology, the Supporting Construction Funds for Shandong Province Data Open Innovation Application Laboratory, the School-level Talent Project (Grant No. 2018RC002), the Weifang Soft Science Project (Grant No. 2023RKX184), and the Weifang City Science and Technology Development Plan Project (Grant No. 2023GX051, 2023 JH14 & 2024GX033).

Data availability

The data can be accessed at <https://data.mendeley.com/datasets/tg3z7xxkdb/1>. The datasets collected and generated in this study are available on request to the corresponding author.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

All authors have consented to the publication of this manuscript.

Competing interests

The authors declare no competing interests.

Received: 26 January 2025 Accepted: 9 May 2025

Published online: 26 May 2025

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